

Aerospace series

**Rectangular bar
Precipitation Hardening Stainless Steel
(X5CrNiCu15-5 (15-5PH))
Thickness $6,0 \text{ mm} \leq a \leq 250,0 \text{ mm}$**

Dimensions

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1 Scope

This standard specifies the dimensions, tolerances and mass of hot wrought flat bars in precipitation hardened stainless steel (X5CrNiCu15-5 (15-5PH)), thickness $6,0 \text{ mm} \leq a \leq 250,0 \text{ mm}$, for aerospace applications.

2 Normative references

This Airbus standard incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any these publications apply to this Airbus standard only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

- | | |
|---------------|---|
| EN3848 | Aerospace series - semi finished metallic products; methods of measuring form deviations ¹ |
| AIMS01-04-000 | Technical specification—Steel wrought products—Plate, sheets, strip, bars and section |
| AIMS01-04-003 | Material specification – Precipitation hardened stainless steel (X5CrNiCu15-5 (15-5PH)) - Solution treated and precipitation hardened - $1070 \text{ MPa} \leq R_m \leq 1200 \text{ MPa}$ – Bar – $6,0 \text{ mm} < a$ or $D \leq 250,0 \text{ mm}$ |

3 Form

See figure 1

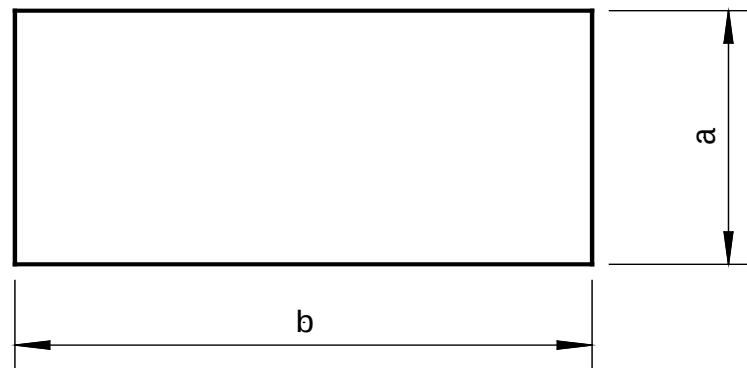


Figure 1:

¹ Published as ASD Prestandard at the date of publication of this standard

4 Recommended dimensions

4.1 Dimensions, cross- sections and mass

See table 1

Table 1: Dimensions

Dimensions in mm				
Size code	a ¹⁾	b ¹⁾	Cross-section (mm2)	Mass ²⁾ per unit length (kg/m)
0612	6	12	72	0,565
0812	8		96	0,754
0616	6	16	96	0,753
0816	8		128	1,004
1016	10		160	1,256
0620	6	20	120	0,942
0820	8		160	1,256
1020	10		200	1,569
1220	12		240	1,883
0625	6	25	150	1,177
0825	8		200	1,569
1025	10		250	1,962
1225	12		300	2,354
0630	6	30	180	1,413
0830	8		240	1,883
1030	10		300	2,354
1230	12		360	2,825
1530	15		450	3,531
1630	16		480	3,767
2030	20		600	4,708
0640	6	40	240	1,883
0840	8		320	2,511
1040	10		400	3,139
1240	12		480	3,767
1540	15		600	4,708
1640	16		640	5,022
2040	20		800	6,278
2540	25		1000	7,847
3040	30		1200	9,417
0650	6	50	300	2,354
0850	8		400	3,139
1050	10		500	3,924
1250	12		600	4,708
1550	15		750	5,885
1650	16		800	6,278
2050	20		1000	7,847
2550	25		1250	9,809
3050	30		1500	11,771
4050	40		2000	15,694
continued				

Table 1: Dimensions

Dimensions in mm				
Size code	a 1)	b 1)	Cross-section (mm2)	Mass 2) per unit length (kg/m)
0655	6	55	330	2,59
0855	8		440	3,453
1055	10		550	4,316
1255	12		660	5,179
1655	16		880	6,906
2055	20		1100	8,632
2255	22		1210	9,495
2555	25		1375	10,79
0860	8	60	480	3,767
1060	10		600	4,708
1560	15		900	7,063
2060	20		1200	9,417
2560	25		1500	11,771
3060	30		1800	14,125
4060	40		2400	18,833
5060	50		3000	23,542
1565	15	65	975	7,651
0670	6	70	420	3,296
0870	8		560	4,394
1070	10		700	5,493
1270	12		840	6,592
1670	16		1120	8,789
2070	20		1400	10,986
2270	22		1540	12,085
2570	25		1750	13,733
2870	28		1960	15,381
3070	30		2100	16,479
3270	32		2240	17,578
3570	35		2450	19,226
4070	40		2800	21,972
5070	50		3500	27,465
6070	60		4200	32,958
3075	30	75	2250	17,656
3275	32		2400	18,833
4075	40		3000	23,542
5075	50		3750	29,427
5575	55		4125	32,37
6075	60		4500	35,313
6575	65		4875	38,255

Continued

Table 1: Dimensions

Dimensions in mm				
Size code	a ¹⁾	b ¹⁾	Cross-section (mm2)	Mass ²⁾ per unit length (kg/m)
0680	6	80	480	3,767
0880	8		640	5,022
0980	9		720	5,65
1080	10		800	6,278
1280	12		960	7,533
1580	15		1200	9,417
1680	16		1280	10,044
2080	20		1600	12,556
2280	22		1760	13,811
2580	25		2000	15,694
2880	28		2240	17,578
3080	30		2400	18,833
3280	32		2560	20,089
3580	35		2800	21,972
4080	40		3200	25,111
5080	50		4000	31,389
5580	55		4400	34,528
6080	60		4800	37,667
6580	65		5200	40,806
7080	70		5600	43,944
1085	10	85	850	6,67
1285	12		1020	8,004
1685	16		1360	10,672
2085	20		1700	13,34
2285	22		1870	14,674
2585	25		2125	16,675
2885	28		2380	18,676
3085	30		2550	20,01
3285	32		2720	21,344
3585	35		2975	23,345
4085	40		3400	26,681
5085	50		4250	33,351
5585	55		4675	36,686
6085	60		5100	40,021
6585	65		5525	43,356
7085	70		5950	46,691
7585	75		6375	50,026
Continued				

Table 1: Dimensions

Dimensions in mm				
Size code	a ¹⁾	b ¹⁾	Cross-section (mm2)	Mass ²⁾ per unit length (kg/m)
0690	6	90	540	4,238
0890	8		720	5,65
1090	10		900	7,063
1290	12		1080	8,475
1690	16		1440	11,3
2090	20		1800	14,125
2290	22		1980	15,538
2590	25		2250	17,656
2890	28		2520	19,775
3090	30		2700	21,188
3290	32		2880	22,6
3590	35		3150	24,719
4090	40		3600	28,25
5090	50		4500	35,313
5590	55		4950	38,844
6090	60		5400	42,375
6590	65		5850	45,906
7090	70		6300	49,438
7590	75		6750	52,969
8090	80		7200	56,5
06100	6	100	600	4,708
08100	8		800	6,278
10100	10		1000	7,847
12100	12		1200	9,417
16100	16		1600	12,556
20100	20		2000	15,694
22100	22		2200	17,264
25100	25		2500	19,618
28100	28		2800	21,972
30100	30		3000	23,542
32100	32		3200	25,111
35100	35		3500	27,465
40100	40		4000	31,389
50100	50		5000	39,236
55100	55		5500	43,16
60100	60		6000	47,083
65100	65		6500	51,007
70100	70		7000	54,931
Continued				

Table 1: Dimensions

Dimensions in mm				
Size code	a ¹⁾	b ¹⁾	Cross-section (mm2)	Mass ²⁾ per unit length (kg/m)
75100	75	100	7500	58,854
80100	80		8000	62,778
85100	85		8500	66,701
06110	6	110	660	5,179
08110	8		880	6,906
10110	10		1100	8,632
12110	12		1320	10,358
16110	16		1760	13,811
20110	20		2200	17,264
22110	22		2420	18,99
25110	25		2750	21,58
28110	28		3080	24,169
32110	32		3520	27,622
35110	35		3850	30,212
40110	40		4400	34,528
50110	50		5500	43,16
70110	70		7700	60,424
06120	6	120	720	5,65
08120	8		960	7,533
10120	10		1200	9,417
12120	12		1440	11,3
16120	16		1920	15,067
20120	20		2400	18,833
22120	22		2640	20,717
25120	25		3000	23,542
28120	28		3360	26,367
30120	30		3600	28,25
32120	32		3840	30,133
35120	35		4200	32,958
40120	40		4800	37,667
50120	50		6000	47,083
60120	60	7200	56,5	
90120	90	10800	84,75	
60130	60	130	7800	61,208
65130	65		8450	66,309
80130	80		10400	81,611
90130	90		11700	91,813
95130	95		12350	96,913
105130	105		13650	107,115
Continued				

Table 1: Dimensions

Dimensions in mm				
Size code	a ¹⁾	b ¹⁾	Cross-section (mm2)	Mass ²⁾ per unit length (kg/m)
06140	6	140	840	6,592
08140	8		1120	8,789
10140	10		1400	10,986
12140	12		1680	13,183
16140	16		2240	17,578
20140	20		2800	21,972
22140	22		3080	24,169
25140	25		3500	27,465
28140	28		3920	30,761
32140	32		4480	35,156
35140	35		4900	38,451
40140	40		5600	43,944
50140	50		7000	54,931
70140	70		9800	76,903
20150	20	150	3000	23,542
30150	30		4500	35,313
40150	40		6000	47,083
60150	60		9000	70,625
110155	110	155	17050	133,795
06160	6	160	960	7,533
08160	8		1280	10,044
10160	10		1600	12,556
12160	12		1920	15,067
16160	16		2560	20,089
20160	20		3200	25,111
22160	22		3520	27,622
25160	25		4000	31,389
28160	28		4480	35,156
32160	32		5120	40,178
35160	35		5600	43,944
40160	40		6400	50,222
50160	50		8000	62,778
70160	70		11200	87,889
80160	80		12800	100,444
06180	6	180	1080	8,475
08180	8		1440	11,3
10180	10		1800	14,125
12180	12		2160	16,95
16180	16		2880	22,6
Continued				

Table 1: Dimensions (concluded)

Dimensions in mm				
Size code	a ¹⁾	b ¹⁾	Cross-section (mm2)	Mass ²⁾ per unit length (kg/m)
20180	20	180	3600	28,25
22180	22		3960	31,075
25180	25		4500	35,313
28180	28		5040	39,55
32180	32		5760	45,2
35180	35		6300	49,438
40180	40		7200	56,5
50180	50		9000	70,625
105180	105		18900	148,313
30190	30	190	5700	44,729
100195	100	195	19500	153,021
20200	20	200	4000	31,389
90200	90		18000	141,25
22210	22	210	4620	36,254
35210	35		7350	57,677
40210	40		8400	65,917
70210	70		14700	115,354
110210	110		23100	181,271
10220	10	220	2200	17,264
40220	40		8800	69,056
70220	70		15400	120,847
80230	80	230	18400	144,389
45250	45	250	11250	88,281
20260	20	260	5200	40,806
100270	100	270	27000	211,875
105270	105		28350	222,469
100280	100	280	28000	219,722
105280	105		29400	230,708
35300	35	300	10500	82,396
10320	10	320	3200	25,111
35345	35	345	12075	94,755
35380		380	13300	104,368
35400		400	14000	109,861
1) Other dimensions on request				
2) For information, calculated with a density of 7,85 kg/dm3				

5 Tolerances

5.1 Dimensional tolerances

5.1.1 Thickness

See table 2.

Table 2: Thickness tolerances

Dimensions in mm

Thickness "a"	Tolerance
≤ 20	± 0,5
20 < a ≤ 40	± 1,0
> 40	± 1,5

5.1.2 Width

See table 3.

Table 3: Width tolerances

Dimensions in mm

Width "b"	Tolerance
≤ 35	± 0,75
35 < w ≤ 75	± 1,0
75 < w ≤ 100	± 1,5
100 < w ≤ 125	± 2,0
> 125	± 2,5

5.1.3 Length

See table 4.

Table 4: Length tolerances

Dimensions in mm

Length type	Length (range)	Length (permissible deviation)
Stock length	3000 to 6000	10% of the quantity delivered may have a minimum length of 1500
Exact length	Up to 6000	+10 0

5.2 Geometrical tolerances

5.2.1 Straightness

5.2.1.1 Method of measurement and symbols

See EN3848.

5.2.1.2 Tolerances

See table 5.

Table 5: Straightness tolerances

Dimensions in mm

Cross-section (mm ²)	Maximum straightness deviation Y ₁ on any length X ₁ ≥ 300
up to 1000	0,004 x l
over 1000	0,0025 x l

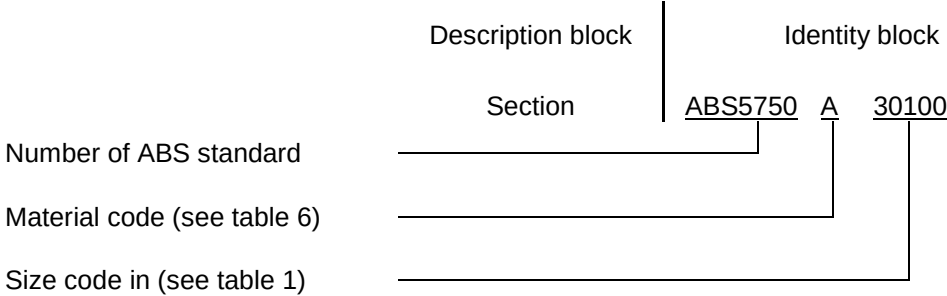
6 Material

See table 6.

Table 6: Material

Material						
delivered				use		
AIMS spec.-no	Tensile strength	condition of treatment	code	AIMS spec.-no	condition of treatment	code
01-04-003	1070 MPa ≤ R _m ≤ 1200 MPa	Solution treated + precipitation hardened ¹⁾	A	01-04-003	Solution treated + precipitation hardened	A
		Solution treated ²⁾	B		Solution treated + precipitation hardened by the end user ¹⁾	
1) solution treated + precipitation hardened = .4 according to DIN designation 2) solution treated = .9 according to DIN designation						

7 Designation



8 Technical specification

See AIMS01-04-000

RECORD OF REVISIONS

Issue	Clause modified	Description of modification
1 07/09	-	New Standard
2 01/10	§4.1	Table 1 Incorporation of additional size code: 100195
3 10/17	§4.1	Table 1 Incorporation of additional size codes : 6070; 1085; 1285;1685; 2085; 2285; 2585; 2885; 3285; 3585; 70110; 90120; 80130; 90130;105130; 110155; 105180; 90200; 22210; 35210; 110210; 40220; 70220; 80230; 100270; 105270; 100280; 105280